ATMEGA328P-AU

- core : 8 bit AVR risc architecture
- Clock speed: up to 20 MHz
- operating voltage: 1.8v to 5.5v
- Package type: TQFP-32 smd
- Maximum current per pin:40 mA
- Total pins: 32 pin

The P stands for power efficient it is lower power consumption MCU

Pinout

Power pins: vcc , gnd, AVCC, AREF

GPIO pins:
23 programmable I/O pins
Each pin has internal pull up resistor
PD0,PD1: UART (RX,TX)
PB3-PB5:SPI
PC4-PC5:I2C(SDA, SCL)
PC0-PC5: ADC input
PB6-PB7: crystal oscillator

Clock system

Internal RC oscillator: 8 MHz
External crystal oscillator: 16MHz with load capacitor

Current consumption

MCU: with 5v at 16MHz external crystal the current
Consumption by MCU = 10 to 20 mA(even when nothing is connected)

Total current consumption= MCU's own current consumption + each external load current
External components do not increase MCU internal current consumption but they draw current through MCU pins so total current consumption in circuit increases

24LC1025 memory

It is EEPROM(electrically erasable programmable read only memory)
Nonvolatile (store the data even after power off)
Total memory:512 K

Pinout
A0, A1, A2 - address select
GND= ground
SDA= I2C data line
SCL= I2C clock line
WP= write protect
VCC= power

- The chip can be connect via I2C
- Each chip is 128KB EEPROM
- Two chip is 256KB total storage
- Operating voltage: 2.5V to 5.5V
- A0, A1, A2: address bits
- vcc= 3.3V
- GND= ground
- SDA , SCL = connect to MCU SDA , SCL via 4.7kohm pull up resistor
- WP= write protect
- If wp= connect with vcc(high) = write disabled and memory is read only
- If wp=connect with gnd (low) = write enabled and memory is readable and writeable
- No pull up resistor required for wp pin , can directly connect with vcc and gnd

A0 , A1, A2 : address pins

- Used to assign different address value when more than one i2c slave and master are connected with same i2c bus line
- Here A0 , A1, A2= connected with vcc = 0x57
- Other A0 , A2= Vcc and A1= gnd = 0x55
- Both memory has different address so master can communicate with particular one by assigning it's specific address to the slave
- Here 2^3= 8 total combination of address range

MCU ATMEGA328P-AU connection

- Pin 3,21,5 = gnd pins = connected to ground
- Pin 4,6 =digital vcc
- Pin 16=AVCC= analog power supply voltage (here in this board both are getting same voltage)
- Battery: through this the whole board will get the power supply
- Power on LED : indication for vcc line to show that board is power on or not
- PB6 and PB7: external crystal connection pins , 16MHz external crystal with load capacitors 22 PF
- Reset: PC6 pin is for reset , connect it with vcc through 10Kohm resistor to keep reset pin high in normal mode
- When MCU goes to reset the MCU pin goes to low

- Don’t keep this reset pin floating otherwise MCU may get randomly reset and power on
- AREF Pin: this pin is used to give stable analog reference voltage
- By this pin the ADC reading will not give wrong value if noise occur
- Keep 0.1 uF capacitor connect with AREF pin to gnd for decoupling
- UART pins: PD0= RX, PD1= TX
- Used for programming and debugging
- SPI interface:
- PB3= MOSI
- PB4=MISO
- PB5=SCK
- I2C interface
- PC4=SDA, PC5 =SCL
- GPIO pins
- PB0-PB5, PD2- PD7, PC0 - PC3